Applications of Statistical Modeling:
Module #1, Lecture #3

Multilevel Modeling Application:
Interviewer Effects in the European Social Survey

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Lecture Overview

- Today, we'll begin with a review of interviewer effects; some of this may be new, and you may have seen some of it in other classes
- We'll then turn to applications involving the European Social Survey (ESS) data, estimating interviewer variance for selected survey variables and interpreting the results
- We'll also discuss how to fit the models using selected software procedures

Interviewer Effects

- Classically viewed as a type of measurement error specific to an interviewer
- Determinants of Intra-Interviewer Correlations ("rho-int"); see West and Blom (2017, JSSAM)
  - Computerization (ACASI/SAQs); correlations expected to be negligible
  - Telephone (smaller) vs. Face-to-face (larger)
  - Items? Open vs. closed, or Factual vs. attitudinal; not many differences
  - Controlled feedback and probing…
  - Interviewer experience / training? Interviewer behaviors / non-verbal signs?
  - Demographics of interviewers and respondents
  - Different recruitment of different types of cases (West and Olson, 2010); argument against measurement error hypothesis

Variance Inflation

- As the sizes of workloads increase, effective sample size is reduced, even for small values of intra-interviewer correlations
- Workload could be reduced by hiring more interviewers, but…
- Wouldn’t that cost more?
- Better to reduce rho-int as much as possible through training, supervision, and good questionnaire design principles

Multilevel Modeling Approaches

- In recent years, methodological studies have taken advantage of multilevel modeling techniques to study interviewer variance
- Effects of individual interviewers j (M_j) on survey measures y for respondent i are treated as **random**:

$$y_{ij} = X_i + M_j + e_{ij}$$

$$M_j \sim N(0, \sigma^2_{\text{int}})$$

$$e_{ij} \sim N(0, \sigma^2)$$

- In most applications, variance in the true values (X_i) and the random errors (e_{ij}) are combined into the error variance term, and a fixed intercept is used

Multilevel Modeling Approaches

- **Advantages:**
  - Capable of handling <u>unbalanced workloads</u> across interviewers (some models require equal workloads!)
  - Capable of including interviewer-specific covariates in an effort to explain variance across interviewers
  - Capable of allowing interviewer variance to vary depending on the type of interviewer (e.g., CAPI vs. CATI)
  - Models with **crossed random effects** enable studies of the contributions of areas and interviewers to overall variance (e.g., O'Muircheartaigh and Campanelli, 1998)
    - Example: interviewers work in multiple randomly sampled PSUs, and we wish to disentangle interviewer and PSU effects

Multilevel Modeling Approaches

- Assuming that interviewers were randomly sampled from some hypothetical super-population of interviewers (the assumption needed to treat effects as random), we can compute "rho-int" (the intra-interviewer correlation) for a given survey measure as follows:

$$\rho_{\text{int}} = \sigma^2_{\text{int}} / \sigma^2_y = \sigma^2_{\text{int}} / (\sigma^2_X + \sigma^2_{\text{int}} + \sigma^2_e) = \sigma^2_{\text{int}} / (\sigma^2_{\text{int}} + \sigma^2_\varepsilon)$$

- We can thus indirectly test whether rho-int is zero by performing appropriate tests for the interviewer variance component (σ^2_{int})

Interviewer Effects in the ESS (Refer to the R Markdown file)

- We will now consider examples of estimating interviewer effects in the European Social Survey (ESS) Data, to illustrate multilevel modeling and diagnostics
- We will begin by using multilevel models to estimate the interviewer variance components for selected variables
- We will then try to explain any interviewer variance by using the interviewer-specific response rates (which could indicate interviewer-specific nonresponse bias)
- We'll focus on R and go through the code and output in great detail; Stata code is on the following slides
- We'll then consider a random coefficient model

Random Intercept Model: Stata Code

```
mixed pplhlp || intnum: , variance reml
* note: mixture-based LRT is automatic!
predict int_eblups, reffects
collapse int_eblups, by(intnum)
qnorm int_eblups, ytitle(EBLUPs of Random
  Interviewer Effects)
predict st_resid, rstandard
qnorm st_resid
* note: add interviewer-level covariate
mixed pplhlp int_rr || intnum: , variance
  reml
```

Random Coefficient Model: Stata Code

```
mixed pplhlp trstplc ||
  intnum: trstplc,
  covariance(unstruct)
  variance reml
```

LRT needs to be done manually. Only automatic for random intercept models.

Three Questions!

Suppose that we have *n* groups/clusters of units:

1. Assuming no other covariates, what are the potential options for modeling the effects of the clusters?
2. What is the difference between an ordinary regression model and a multilevel regression model?
3. What if we have covariates both for Level 1 and Level 2?

Ordinary versus Multilevel Regression

- Ordinary regression: a reference category and *n*-1 indicators
  - Mean structure changed
- Multilevel regression: *n* indicators
  - Variance structure changed
  - Better for many groups and small sizes
  - Flexible to expand
  - Borrow information across groups: **partial pooling**

Partial Pooling without Predictors:

- Complete pooling—excluding a group indicator $y_{ij} = \mu_0 + \epsilon_{ij}$
- No pooling—estimating separate models with each group $j = 1, \ldots, n$, $y_{ij} = \mu_{0j} + \epsilon_{ij}$
- Partial pooling-multilevel model: $y_{ij} = \mu_{0j} + \epsilon_{ij}, \mu_{0j} \sim N(0, \sigma_0^2)$

Partial-pooling Estimates

- Weighted average between unpooled estimate $\bar{y}_j$ and the completely pooled estimate $\bar{y}_{all}$
- Capture unit and group-level variation

$$\hat{\mu}_{0j}^{multilevel} \approx \frac{\frac{n_j}{\sigma^2}\bar{y}_j + \frac{1}{\sigma_0^2}\bar{y}_{all}}{\frac{n_j}{\sigma^2} + \frac{1}{\sigma_0^2}}$$

- when the group size n_j is large, or the between-group variance σ_0^2 is large, $\hat{\mu}_{0j}^{multilevel}$ will be close to $\bar{y}_j$
- when the group size n_j is small, or the between-group variance σ_0^2 is small, $\hat{u}_{0j}^{multilevel}$ will be close to $\bar{y}_{all}$

Adding Predictors

- Multilevel model with random intercepts and slopes:

$$y_{ij} = \alpha_i + \beta_i x_{ij} + \epsilon_{ij}$$

$$\alpha_i = \gamma_0^\alpha + \gamma_1^\alpha u_i + \gamma_i^\alpha$$

$$\beta_i = \gamma_0^\beta + \gamma_1^\beta u_i + \gamma_i^\beta$$

- Combine as:

$$y_{ij} = \gamma_0^\alpha + \gamma_1^\alpha u_i + \gamma_0^\beta x_{ij} + \gamma_1^\beta u_i x_{ij} + \gamma_i^\alpha + \gamma_i^\beta x_{ij} + \epsilon_{ij}$$

Panel Surveys with Complex Sample Designs

- Consider panel surveys with a complex *initial* sample designs (stratification, cluster sampling, weighting)
- Repeatedly measure the initially sampled units at multiple follow-up waves
- Next week, we will introduce design-based and model-based approaches to account for:
  - Correlation of repeated measurements
  - Complex sampling features
  - Different patterns of nonresponse

Some Final Thoughts

- Importantly, we would make the same inferences regardless of the software used
- "Automatic" mixture-based LRTs of variance components and intra-interviewer correlations are generally available, <u>for random intercept models</u>
- Some amount of work may be required for diagnostics in each package, **but it is important to always consider them**

Your Analysis Projects

- **Be creative!** Do some data exploration, and try to find some interesting patterns in your data…
- Try a random coefficient model! Not a lot of research considers between-cluster variance in regression coefficients…important for longitudinal data
- Try to explain variance in the residuals and the random effects! Consider aggregating L1 variables…
- Carefully consider model diagnostics!
- Be <u>**careful**</u> and <u>**clear**</u> with your interpretations